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The Mean Value Inequality

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Theorem 1. *polygonal_connected_of_connected_lem : Visopenlem : Viscloseddef : polygonalLineif Uisanopenco*

Proof. □

Lemma 2. *def:polygonalLine let $x \in U$, and V the set of all points in U that can be joined to x by a polygonal line contained in U . Then V is open.*

Proof. let $a \in V$ be the end of the polygonal line l which starts from x . Since U is open, there exist a ball $B(a, r)$ contained in U . Every points $p \in B(a, r)$ can be joined to a by a straight line. Therefore, by joining the two polygonal line, we have a polygonal line joining x and p . Hence, $B(a, r) \subset V$ and V is open. □

Lemma 3. *def:polygonalLine V is closed in U .*

Proof. Let $a \in \overline{V}$, Since U is open, there exists a ball $B(a, r)$ contained in U . Since a is in the closure of V , there exists a point $b \in B(a, r) \cap V$. Then a can be joined to b via a straight line contained in U and b can be joined to x by a polygonal line in U . Therefore, a can be joined to x by a polygonal line in U . Hence $a \in V$ and V is closed in U . □

Proof of Theorem ??. Since V is both open and closed in U , and U is connected then either $V = U$ or $V = \emptyset$. If U is not empty then V contained x and is not empty, Hence $V = U$.

Otherwise if $U = \emptyset$, then the result is trivial. □

Definition 4. *thm:existence of polygonal line Let U be an open connected set of a Banach space E . for $a, b \in U$, define $d_U(a, b)$ to be the infimum of the lengths of the polygonal lines contained in U joining a and b .*

Theorem 5. *def:distance in U $d_U(a, b)$ is a distance in the topological space U .*

Proof. Let $\varphi: [0, 1] \rightarrow U$ be an arbitrary polygonal line joining a to b in U , and $t_0, \dots, t_n \in [0, 1]$ the finitely many points such that $\varphi: [t_i, t_{i+1}] \rightarrow U$ is affine for all $i \in \{0, \dots, n-1\}$.

Positivity: The length of φ is $L((t_i), n, \varphi) := \sum_{i=0}^{n-1} \|\varphi(t_{i+1}) - \varphi(t_i)\|$ which by the definition of a norm function is strictly non-negative. Since $d_U(a, b)$ is the infimum of a strictly non-negative set it must itself be non-negative. If $d_U(a, b) = 0$, by a previous theorem $\|b - a\| \leq d_U(a, b) = 0$, which implies $a = b$.

Symmetry: For any polygonal line φ from a to b in U , the reverse path $\psi: [0, 1] \rightarrow U$ defined $\psi(x) = \varphi(1-x)$ is a polygonal line from b to a . $\text{Length}(\psi) = \sum_{i=0}^{n-1} \|\psi(1-t_i) - \psi(1-t_{i+1})\| = \sum_{i=0}^{n-1} \|\varphi(t_{i+1}) - \varphi(t_i)\| = \text{Length}(\varphi)$, so the set of lengths of polygonal lines from a to b and from b to a are equivalent and therefore $d_U(a, b) = d_U(b, a)$.

Triangle Inequality: Let $\varphi: [0, 1] \rightarrow U$ be a polygonal line from a to b and $\psi: [0, 1] \rightarrow U$ be a polygonal line from b to c . The extension of φ by ψ is a polygonal line $\theta: [0, 1] \rightarrow U$ from a to c , where

$$\theta(x) = \begin{cases} \varphi(2x) & \text{if } x \in [0, \frac{1}{2}] \\ \psi((2x-1)) & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

By the definition of the infimum, $d_U(a, c) \leq \text{Length}(\theta) = \text{Length}(\varphi) + \text{Length}(\psi)$, and since φ and ψ are arbitrary, $d_U(a, c) \leq d_U(a, b) + d_U(b, c)$. $\square$

Theorem 6. *def:distance in U thm:existence of polygonal line norm_image_subie_of_norm_h as F Deriv Within l e' Let $U \rightarrow F$ be a differentiable mapping with values in a Banach space F . Assume that*

$$\|f'(x)\| \leq k, \forall x \in U$$

Then, for any $x_1, x_2 \in U$, $\|f(x_1) - f(x_2)\| \leq k \cdot d_U(x_1, x_2)$.

Proof. Let $x_1, x_2 \in U$ and l a polygonal line contained in U joining x_1 and x_2 . then let l be composed of n line segments with endpoints $(a_0, a_1), (a_1, a_2), \dots, (a_{n-1}, a_n)$, where $a_0 = x_1, a_n = x_2$ and $a_i \in U$. Then from the mean value inequality on each interval (a_i, a_{i+1}) , we have $\|f(a_i) - f(a_{i+1})\| \leq k \cdot \|a_i - a_{i+1}\|$. From the triangle inequality we have

$$\|f(x_1) - f(x_2)\| \leq \sum_{i=0}^{n-1} \|f(a_i) - f(a_{i+1})\| \leq \sum_{i=0}^{n-1} k \cdot \|a_i - a_{i+1}\| = k \cdot \text{length}(l)$$

as $\text{length}(l)$ is the sum of the length of each interval. Therefore, taking the infimum we get,

$$\|f(x_1) - f(x_2)\| \leq k \cdot \inf \text{length}(l) = k \cdot d_U(x_1, x_2)$$

$\square$

Definition 7. In a topological space X a continuous mapping $\varphi: [0, 1] \rightarrow X$ is called a *path*; the point $\varphi(0)$ is called the *origin of the path* and the point $\varphi(1)$ is called the *end of the path*. Two points $a, b \in X$ can be joined by a path if there exists φ such that $\varphi(0) = a$ and $\varphi(1) = b$

Definition 8. Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$, with subset $A \subseteq E$. A *polygonal line* in A is a path $\varphi: [0, 1] \rightarrow A$ such that there exists $n \geq 1$ with points $t_0, \dots, t_n$ on the interval $[0, 1]$:

$$t_0 = 0 < t_1 < \dots < t_{n-1} < t_n = 1$$

so that for each $i \in \{0, \dots, n-1\}$, the restriction of φ on the interval $[t_i, t_{i+1}]$ is an affine map. This is to say that $t \in [t_i, t_{i+1}] \implies \varphi(t) = t \cdot v_i + c_i$ for some $v_i, c_i \in E$. Hence the image of $[t_i, t_{i+1}]$ in φ is a straight-line segment in A from $\varphi(t_i)$ to $\varphi(t_{i+1})$.

Definition 9. Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. The *length of an interval* with origin $a \in E$ and end $b \in E$ is $d(a, b) := \|b - a\|$.

Definition 10. Given a partition $(t_i)_{i=0}^n$ for some fixed $n \geq 1$, the *length of a polygonal line* φ respecting this partition is the sum of the lengths of the intervals from which it is formed:

$$L((t_i), n, \varphi) := \sum_{i=0}^{n-1} \|\varphi(t_{i+1}) - \varphi(t_i)\|$$

Lemma 11. *Given a sequence $(a_i)_{i=0}^n$ with $a_i \in E$ for some $n \geq 1$, $\sum_{i=0}^{n-1} \|a_{i+1} - a_i\| \geq \|a_n - a_0\|$*

Proof of Lemma ??. • **Base Case ($n = 1$):** For $n = 1$, we need $\|a_1 - a_0\| \geq \|a_1 - a_0\|$ which follows immediately by reflexivity.

- **Induction Hypothesis:** Assume that the result holds for any such sequence $(b_i)_{i=0}^k$. That is,

$$\sum_{i=0}^{k-1} \|b_{i+1} - b_i\| \geq \|b_k - b_0\|$$

- **Induction Step:** We want to show that $\sum_{i=0}^k \|a_{i+1} - a_i\| \geq \|a_{k+1} - a_0\|$ must follow from this assumption, for some arbitrary sequence $(a_i)_{i=0}^{k+1}$.

$$\begin{aligned} \|a_{k+1} - a_0\| &= \|(a_{k+1} - a_k) + (a_k - a_0)\| \\ &\leq \|a_{k+1} - a_k\| + \|a_k - a_0\| \\ &\leq \|a_{k+1} - a_k\| + \sum_{i=0}^{k-1} \|a_{i+1} - a_i\| \\ &= \sum_{i=0}^k \|a_{i+1} - a_i\| \end{aligned}$$

Therefore, by the principle of mathematical induction, $\sum_{i=0}^{n-1} \|a_{i+1} - a_i\| \geq \|a_n - a_0\|$ for any such sequence with any $n \geq 1$. $\square$

Theorem 12. *The length of a polygonal line φ respecting a partition $(t_i)_{i=0}^n$ for some fixed $n \geq 1$ between origin $a = \varphi(0)$ and end $b = \varphi(1)$ is at least equal to the length of the interval between these points:*

$$\sum_{i=0}^{n-1} \|\varphi(t_{i+1}) - \varphi(t_i)\| \geq \|\varphi(1) - \varphi(0)\|$$

Proof of Theorem ??. We consider the sequence of points $(p_i)_{i=0}^n$ with $p_i := \varphi(t_i)$. Then by the previous lemma, with $p_0 = \varphi(0)$ and $p_n = \varphi(1)$, our desired result follows immediately. $\square$